COER UNIVERSITY

DBMS LAB ASSIGNMENT | LAB SHEET 3: SQL FUNCTIONS

Section A: Number Functions

Question 1

Question: Demonstrate ABS() with suitable examples.

Answer:

```
SELECT ABS(-25.5) AS AbsVal, ABS(10) AS PosVal;
```

Explanation: ABS(x) returns the positive absolute value of a number.

Question 2

Question: Demonstrate ROUND() with suitable examples.

Answer:

```
SELECT ROUND(123.4567, 2) AS RoundedTwoDec, ROUND(123.4567, 0) AS RoundedInt;
```

Explanation: ROUND(x, d) rounds x to d decimal places.

Question 3

Question: Demonstrate CEIL() and FLOOR().

Answer:

```
SELECT CEIL(12.3) AS CeilVal, FLOOR(12.8) AS FloorVal;
```

Explanation: CEIL() returns the smallest integer >= x; FLOOR() returns the largest integer <= x.

Question 4

Question: Demonstrate MOD() with suitable examples.

Answer:

```
SELECT MOD(17, 5) AS Remainder, 17 % 5 AS AltRemainder;
```

Explanation: MOD(N, M) returns the remainder of N divided by M.

Question 5

Question: Calculate the rounded salary of employees.

Answer:

```
SELECT Name, Salary, ROUND(Salary, 0) AS RoundedSalary FROM Employee;
```

Explanation: Applies ROUND() to the Salary column.

Section B: Aggregate Functions

Question 6

Question: Find the total number of students using COUNT().

Answer:

```
SELECT COUNT(*) AS TotalStudents FROM Student;
```

Explanation: COUNT(*) returns the total number of rows in the table.

Question 7

Question: Find the total salary of all employees using SUM().

Answer:

```
SELECT SUM(Salary) AS TotalSalary FROM Employee;
```

Explanation: SUM() calculates the sum of all values in a numeric column.

Question 8

Question: Find the average salary using AVG().

Answer:

```
SELECT AVG(Salary) AS AvgSalary FROM Employee;
```

Explanation: AVG() calculates the mean value of a numeric column.

Question 9

Question: Find the highest salary using MAX().

Answer:

```
SELECT MAX(Salary) AS HighestSalary FROM Employee;
```

Explanation: MAX() retrieves the maximum value from a column.

Question 10

Question: Find the lowest salary using MIN().

Answer:

```
SELECT MIN(Salary) AS LowestSalary FROM Employee;
```

Explanation: MIN() retrieves the minimum value from a column.

Section C: Character Functions

Question 11

Question: Demonstrate UPPER() on student names.

Answer:

```
SELECT Name, UPPER(Name) AS UpperName FROM Student;
```

Explanation: UPPER() converts text to uppercase.

Question 12

Question: Demonstrate LOWER() on student names.

Answer:

```
SELECT Name, LOWER(Name) AS LowerName FROM Student;
```

Explanation: LOWER() converts text to lowercase.

Question 13

Question: Demonstrate LENGTH() on student names.

Answer:

```
SELECT Name, LENGTH(Name) AS NameLength FROM Student;
```

Explanation: LENGTH() returns the character length of string values.

Question 14

Question: Demonstrate CONCAT() using first and last names.

Answer:

```
SELECT CONCAT('Aman', ' ', 'Sharma') AS FullName;
```

Explanation: CONCAT() combines multiple string arguments into a single string.

Question 15

Question: Demonstrate SUBSTRING()/SUBSTR() with suitable examples.

Answer:

```
SELECT SUBSTRING('Database Management', 1, 8) AS SubStrVal;
```

Explanation: SUBSTRING(str, pos, len) extracts a portion of text starting at pos for length len.

Section D: Conversion Functions

Question 16

Question: Convert a number into a character/string value.

Answer:

```
SELECT CAST(12345 AS CHAR) AS StrValue;
```

Explanation: CAST() converts a numeric constant into string data type CHAR.

Question 17

Question: Convert a string into a numeric value.

Answer:

```
SELECT CAST('987.65' AS DECIMAL(10,2)) AS NumValue;
```

Explanation: CAST() parses a numeric string into a DECIMAL number.

Question 18

Question: Convert a date value into a different format.

Answer:

```
SELECT DATE_FORMAT(CURDATE(), '%d-%b-%Y') AS FormattedDate;
```

Explanation: DATE_FORMAT() formats a date according to specifiers (e.g. 24-Sep-2026).

Question 19

Question: Demonstrate CAST() with suitable examples.

Answer:

```
SELECT CAST('2026-09-24' AS DATE) AS ConvertedDate;
```

Explanation: Converts string literal into SQL DATE data type.

Question 20

Question: Demonstrate CONVERT() where supported by the DBMS.

Answer:

```
SELECT CONVERT('100', SIGNED) AS SignedInt;
```

Explanation: CONVERT() converts an expression to a target data type in MySQL.

Section E: Date Functions

Question 21

Question: Display the current date using a suitable date function.

Answer:

```
SELECT CURDATE() AS TodayDate, CURRENT_DATE() AS CurrentDate;
```

Explanation: CURDATE() returns the current date from the operating system.

Question 22

Question: Extract the year, month and day from a date.

Answer:

```
SELECT YEAR(CURDATE()) AS YR, MONTH(CURDATE()) AS MTH, DAY(CURDATE()) AS DY;
```

Explanation: YEAR(), MONTH(), and DAY() extract specific date components.

Question 23

Question: Find the difference between two dates.

Answer:

```
SELECT DATEDIFF('2026-12-31', '2026-09-24') AS DaysDiff;
```

Explanation: DATEDIFF(expr1, expr2) calculates expr1 - expr2 in days.

Question 24

Question: Add a specified number of days to a date.

Answer:

```
SELECT DATE_ADD('2026-09-24', INTERVAL 30 DAY) AS FutureDate;
```

Explanation: DATE_ADD() adds an interval (30 days) to a starting date.

Question 25

Question: Display employees' joining dates in a required date format.

Answer:

```
SELECT Name, DATE_FORMAT('2025-08-01', '%W, %M %d, %Y') AS FormattedJoining FROM Employee
LIMIT 1;
```

Explanation: Applies DATE_FORMAT() to display joining dates in verbose format.